1. ML workflow — short notes

1. **Define the problem**
    - Identify the input features XX and target yy.
    - Regression → predict a number.
    - Classification → predict a category/class.
2. **Understand the data**
    - Look at columns, data types, distributions, and basic statistics.
    - Identify numerical and categorical features.
3. **Preprocess the data**
    - Handle missing values.
    - Remove duplicates.
    - Fix typos/inconsistent categories.
    - Correct incorrect data types.
    - Encode categorical variables.
    - Check for unreasonable values/outliers.
4. **Exploratory Data Analysis**
    - Look for relationships between variables.
    - Correlations.
    - Distributions.
    - Useful plots.
5. **Split the data**
    - Separate features and target.
    - Create training and testing sets.
    - Avoid data leakage.
6. **Train the model**
    - Fit the model using the training data.
7. **Evaluate**
    - Linear regression: MAE, RMSE, R2R^2.
    - Logistic regression: accuracy, precision, recall, F1-score, confusion matrix.
8. **Improve**
    - Try better features.
    - Remove irrelevant features.
    - Compare preprocessing choices.
    - Tune the model.

TASK

Use the provided housing dataset.

Part A — Linear Regression

Predict `sale_price_eur`.

The student should:

- inspect and clean the dataset;
- explain every preprocessing decision;
- choose suitable features;
- encode categorical variables;
- split into training and test sets;
- train a linear regression model;
- evaluate using MAE, RMSE and R2R^2;
- interpret which features appear to affect house prices most.

Part B — Logistic Regression

Predict `high_price`, where:

high_price={1house price≥€500,0000otherwise\text{high_price} = \begin{cases} 1 & \text{house price} \ge €500,000\\ 0 & \text{otherwise} \end{cases}

Train a logistic regression classifier and evaluate it using accuracy, precision, recall, F1-score and a confusion matrix.

Important: **do not use `sale_price_eur` as an input feature when predicting `high_price`**. That would be data leakage. `house_id` should also not be used as a predictive feature.

The dataset contains **1,208 houses and 19 columns**, with deliberate missing values, inconsistent spelling/capitalization, malformed numerical values, categorical inconsistencies and duplicate records.